

To determine the maximum possible value of g across all valid choices of $a_1, \dots, a_k$ with $k \geq 3$, we analyze the conditions imposed by the problem. The key requirement is that g divides each $\left(\frac{P}{a_i} + 1\right)$ for all i , where $P = a_1 a_2 \cdots a_k$. This implies $\frac{P}{a_i} \equiv -1 \pmod{g}$ for each i , leading to the congruence $P \equiv -a_i \pmod{g}$ for all i . Consequently, all a_i are congruent modulo g , say to r , and $r \equiv -P \pmod{g}$. Substituting $P \equiv r^k \pmod{g}$ yields $r \equiv -r^k \pmod{g}$, or $r(1 + r^{k-1}) \equiv 0 \pmod{g}$. Since $r \not\equiv 0 \pmod{g}$, we require $1 + r^{k-1} \equiv 0 \pmod{g}$.

To maximize g , we consider $k = 3$, as larger k would restrict g further due to the sum constraint. For $k = 3$, the condition becomes $r^2 \equiv -1 \pmod{g}$, necessitating that -1 is a quadratic residue modulo g . This requires g to be a product of primes $\equiv 1 \pmod{4}$ (or 2, but 2 cannot contribute significantly). We seek the largest $g < 5 \times 10^5$ (to ensure the sum $3(r + g) < 3 \times 10^6$) composed of such primes.

The primes $\equiv 1 \pmod{4}$ less than 5×10^5 include 5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, etc. Testing combinations, the product $5 \times 17 \times 61 \times 97 = 502,945$ is the largest under 5×10^5 . We verify -1 is a quadratic residue modulo each prime factor and construct r via the Chinese Remainder Theorem. For $g = 502,945$, choosing $r \equiv 414,362 \pmod{502,945}$ satisfies $r^2 \equiv -1 \pmod{g}$. The numbers $a_1 = 414,362$, $a_2 = 917,307$, $a_3 = 1,420,252$ have sum $2,751,921 < 3 \times 10^6$, and each $\left(\frac{P}{a_i} + 1\right)$ is divisible by 502,945.

No larger g meets all conditions. Thus, the maximum g is 502945.